

AP Calc Lesson 1.1 Homework

Answer
Key

1. a. $\frac{34 \text{ miles}}{1.5 \text{ hours}} = 22\frac{2}{3}$ miles per hour
b. It is slower than his instantaneous speed. At 9 AM he was riding faster than average.
c. The driver's speedometer gives the current speed, or the rate at which he is riding at one particular instant, which is called the instantaneous rate of change.
2. a. $\frac{H(11) - H(0)}{11 - 0} = \frac{10.2 - 12.0}{11} = -\frac{1.8}{11} = -0.164 \text{ cm/minute}$
b. $\approx \frac{H(7) - H(3)}{7 - 3} = \frac{10.9 - 11.6}{4} = -\frac{0.7}{4} = -0.175 \text{ cm/minute}$
3. Dwaina is correct. We cannot use the standard difference quotient to determine the instantaneous rate of change for the reasons Jenny and Kyle identified. We can estimate the rate by choosing a small interval surrounding 6 AM and calculating the average rate of change over that very short interval, as Dwaina suggests.
4. a. $\frac{C(8) - C(1)}{8 - 1} = \frac{14}{7} = 2$ customers per hour
b. $\frac{C(3.5) - C(3)}{3.5 - 3} = 7$; $\frac{C(3.1) - C(3)}{3.1 - 3} = 7.8$; $\frac{C(3.01) - C(3)}{3.01 - 3} = 7.98$; $\frac{C(3.001) - C(3)}{3.001 - 3} = 7.998$
c. The average rate of change values are getting closer and closer to 8 customers per hour.
5. a. Greater at $x = 2$ because the slope of f is positive at both x -values but steeper at $x = 2$ than at $x = 3$.
b. $x = 0$ or $x = 4$
c. Sample answers: $[0, 6]$, $[-2, 4]$
6. a. $[40, 60]$; each time interval is 20 seconds so the denominator of each average rate of change calculation is the same. Thus, we can focus only on the numerator, which represents the change in data downloaded over the 20-second period. The change in data downloaded is greatest between $t = 40$ and $t = 60$ ($580 - 360 = 220$ Mb).
b. Faster; the average download speed on $[20, 80]$ is $\frac{700 - 150}{60} = 9.167 \text{ Mb/second}$ so the current download speed at $t = 60$ of 14 Mb/second is faster.
7. a. $-\frac{5}{2}$ meters/minute
b. At $t = 5$, the robot is moving toward the charging station at a rate of $\frac{5}{2}$ meters per minute.

8. V, I, IV, III, II

9. C

10. B

AP Calc Lesson 1.2 Homework

Answer Key

1. a. $\lim_{x \rightarrow 3} f(x) = -5$
b. $\lim_{x \rightarrow -2^+} g(x) = 10$
c. $\lim_{x \rightarrow 8^-} h(x) = 0$

2. a. -3
b. 0
c. 0
d. 0
e. 2
f. 0
g. Does not exist
h. 2
i. 3

3. 7

4. $\lim_{x \rightarrow 3^-} g(x) = 2 \sin\left(\frac{3\pi}{2}\right) - 3 = -2 - 3 = -5$
 $\lim_{x \rightarrow 3^+} g(x) = 3^2 - 4(3) + 2 = -1$
Since $\lim_{x \rightarrow 3^-} g(x) \neq \lim_{x \rightarrow 3^+} g(x)$, $\lim_{x \rightarrow 3} g(x)$ does not exist

5. Answers will vary. A correct answer will have a y -intercept of -15 and an x -intercept at $x = -5$. Sample answer: $f(x) = (x + 5)(x - 3)$

6. a. Guaranteed to be true.
b. Might be true.
c. Guaranteed to be false.
d. Guaranteed to be true.
e. Might be true.

7. a. 12

b. Does not exist; $\lim_{x \rightarrow 4^-} \frac{1}{x-4} = -\infty$ and $\lim_{x \rightarrow 4^+} \frac{1}{x-4} = \infty$

c. 2

8. a. $b = 10$

b. Many answers possible as long as $b \neq 10$.

9. D

10. B

AP Calc Lesson 1.3 Homework

Answer Key

1. -1

2. $\frac{2}{7}$

3. a. $\lim_{x \rightarrow 1} g(x) = 2$

b. $\lim_{x \rightarrow -1^-} g(x) = -\infty$; $\lim_{x \rightarrow -1^+} g(x) = \infty$

4. $a = 2, a = 5$

5. $a = 3, b = -5, c = 3$

6. One vertical asymptote at $x = -3$.

$$g(x) = \frac{x-3}{(x^2-9)(x^2+9)} = \frac{x-3}{(x-3)(x+3)(x^2+9)}$$

The graph of g has a hole at $x = 3$ and vertical asymptotes at $x = -3$. $x^2 + 9 = 0$ has no solutions over the real numbers.

7. $a = 16, b = 7$

8. Answers may vary. Sample answers: $f(x) = \frac{16x(x-8)}{x(x+4)}$ or $f(x) = \frac{4x(x-8)(x+4)}{x(x+4)}$

9. C

10. B

AP Calc Lesson 1.4 Homework

Answer Key

1. a. 0
 b. Does not exist (unbounded)
2. As the distance from the highway continues to increase, the traffic noise does not decrease past 45 decibels.
3. 0
4. $\frac{8}{5}$
5. Answers will vary. Any 4th degree polynomial with a leading coefficient of -6 is correct.
6. A correct answer will have equal degree polynomials in the numerator and denominator where the ratio of leading coefficients is 2. The denominator will have a factor of $x + 3$. The numerator will have a factor of $x - 4$. Sample answer: $g(x) = \frac{2x - 8}{x + 3}$
7. $y = -\frac{2}{3}$
8. $a = -5, b = 40, c = -10, d = 16$
9. A
10. C

AP Calc Lesson 1.5 Homework

Answer Key

1. a. -1

b. 1

c. DNE

2. a. 3

b. $\lim_{x \rightarrow 2} (7g(x) - f(x)) = 7 \lim_{x \rightarrow 2} g(x) - \lim_{x \rightarrow 2} f(x) = 7(4) - 3 = 25$

3. $\lim_{x \rightarrow 4} \frac{x^2 - 16}{\sqrt{x} - 2} = \lim_{x \rightarrow 4} \frac{x^2 - 16}{\sqrt{x} - 2} \cdot \frac{\sqrt{x} + 2}{\sqrt{x} + 2} = \lim_{x \rightarrow 4} \frac{(x^2 - 16)(\sqrt{x} + 2)}{x - 4} = \lim_{x \rightarrow 4} (x + 4)(\sqrt{x} + 2) = 8 \cdot 4 = 32$

4. $\lim_{x \rightarrow 2^+} [-5f(x - 2)] = -5 \lim_{x \rightarrow 2^+} \frac{|x - 2|}{x - 2} = -5$

5. a. $\lim_{x \rightarrow -1^+} \frac{f(x)}{4} = \frac{1}{4} \lim_{x \rightarrow -1^+} f(x) = \frac{3}{2}$

b. $\lim_{x \rightarrow -1} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow -1} f(x)}{\lim_{x \rightarrow -1} g(x)} = -3$

c. $\lim_{x \rightarrow -1} [f(x) \cdot g(x)] = \lim_{x \rightarrow -1} f(x) \cdot \lim_{x \rightarrow -1} g(x) = -12$

6. $\lim_{x \rightarrow 0} \frac{\sin(kx)}{4x} = \frac{k}{4} = 4 \rightarrow k = 16$

7. a. $\lim_{x \rightarrow 0} [2g(x + 1)] = 2 \lim_{x \rightarrow 0} g(x + 1) = 6$

b. $\lim_{x \rightarrow 0^-} [f(x) - g(x)] = \lim_{x \rightarrow 0^-} f(x) - \lim_{x \rightarrow 0^-} g(x) = -4 - 3 = -7$

$$\lim_{x \rightarrow 0^+} [f(x) - g(x)] = \lim_{x \rightarrow 0^+} f(x) - \lim_{x \rightarrow 0^+} g(x) = -3 - 4 = -7$$

$$\lim_{x \rightarrow 0} [f(x) - g(x)] = -7$$

c. $\lim_{x \rightarrow 2} g(x) = 2; f(2) = 3; \lim_{x \rightarrow 2} [f(g(x))] = 3$

8. $\lim_{x \rightarrow 0} \frac{\tan(4x)}{3x \sec(6x)} = \lim_{x \rightarrow 0} \frac{\sin(4x)}{3x} \cdot \frac{\cos(6x)}{\cos(4x)} = \frac{4}{3}$

9. C

10. C

AP Calc Lesson 1.6 Homework

Answer Key

1. $\lim_{x \rightarrow 0} f(x) = 2$
2. $\lim_{x \rightarrow 3} g(x) = 9$ and $\lim_{x \rightarrow 3} h(x) = 10$. Since $9 \neq 10$, $\lim_{x \rightarrow 3} f(x)$ cannot be determined using the given information.
3. a. Answers may vary. Example: $g(x) = -x^2 - 2$
b. Answers may vary. Example: $h(x) = |x| - 2$
4. $\lim_{x \rightarrow 2} g(x) = \lim_{x \rightarrow 2} h(x) = 3$ so, using the squeeze theorem, $\lim_{x \rightarrow 2} f(x) = 3$
5. $\lim_{x \rightarrow 4} g(x) = \lim_{x \rightarrow 4} h(x) = 6$ so, using the squeeze theorem, $\lim_{x \rightarrow 4} f(x) = 6$
6. a. $-1 \leq \cos(x) \leq 1$
b. $1 \leq 2 - \cos(x) \leq 3$
c. $g(x) = \frac{1}{x+3}$ and $h(x) = \frac{3}{x+3}$
d. $\lim_{x \rightarrow \infty} \frac{2 - \cos(x)}{x+3} = 0$
7. $-x^4 - 2 \leq x^4 \sin\left(\frac{6}{x}\right) - 2 \leq x^4 - 2$ and $\lim_{x \rightarrow 0} (-x^4 - 2) = \lim_{x \rightarrow 0} (x^4 - 2) = -2$, so using the Squeeze Theorem $\lim_{x \rightarrow 0} \left[x^4 \sin\left(\frac{6}{x}\right) - 2 \right] = -2$
8. C
9. D

AP Calc Lesson 1.7 Homework

Answer Key

1.
 - a. $x = -6$ and $x = -2$
 - b. $x = -4$
 - c. $(-\infty, -6), (-6, -4), (-4, -2)$, and $(-2, \infty)$
2. $f(x) = \frac{(x-2)(x-6)}{(x-2)(x+5)}; x-2=0 \rightarrow x=2$
3. $x = -3$ removable discontinuity, $x = 0$ infinite discontinuity, $x = 2$ removable discontinuity, $x = 4$ jump discontinuity
4. No, this only guarantees that $\lim_{x \rightarrow 2} g(x) = 8$. For g to be continuous, it must also be true that $g(2) = 8$.
5. $\lim_{x \rightarrow 2^-} f(x) = 4 \cos(\pi \cdot 2) = 4(1) = 4$
 $\lim_{x \rightarrow 2^+} f(x) = (2)^2 - \ln(2-1) = 4 - \ln(1) = 4 - 0 = 4$
 $f(2) = (2)^2 - \ln(2-1) = 4 - \ln(1) = 4 - 0 = 4$
Since $\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^+} f(x) = 4$ and $\lim_{x \rightarrow 2} f(x) = f(2) = 4$, f is continuous at $x = 2$.
6.
 - a. $(-\infty, -7)$ and $(-7, \infty)$
 - b. $x = -7$; $\lim_{x \rightarrow -7^-} f(x) = -1$ but $\lim_{x \rightarrow -7^+} f(x) = 1$ therefore f has a jump discontinuity at $x = -7$.
 $(-1 \neq 1)$.
7. $\lim_{x \rightarrow 2^-} f(x) = 3(2)^2 - 5(2) + 1 = 12 - 10 + 1 = 3$
 $\lim_{x \rightarrow 2^+} f(x) = \lim_{x \rightarrow 2^+} g(x) = 5$
 f is not continuous at $x = 2$ as $\lim_{x \rightarrow 2^-} f(x) \neq \lim_{x \rightarrow 2^+} f(x)$
8. $a = 4, b = -6, c = 8$
 $ax - 8 = 0$ when $x = 2$; $2a - 8 = 0 \rightarrow a = 4$
 $x^2 + bx + c = 0$ when $x = 2$; $4 + 2b + c = 0$
 $x^2 + bx + c = 0$ when $x = 4$; $16 + 4b + c = 0$
 $4 + 2b + c = 16 + 4b + c \rightarrow 2b = -12 \rightarrow b = -6$
 $4 + 2(-6) + 0 \rightarrow c = 8$

9. C

10. C

AP Calc Lesson 1.8 Homework

Answer Key

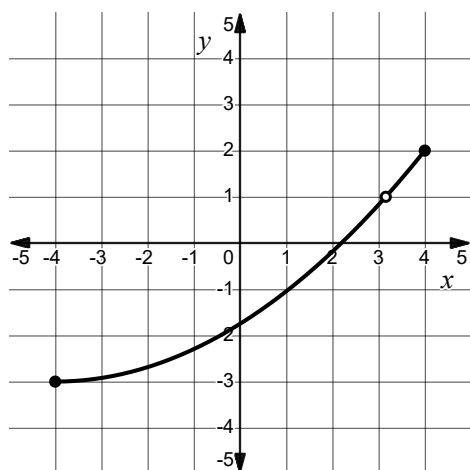
1. Benny is correct. f has a removable discontinuity at $x = -3$ and this discontinuity could be removed by redefining $f(-3)$ so that $f(-3) = \lim_{x \rightarrow -3} f(x) = -2$.
2. $k = -12$
3. $x = 1$, non-removable
 $x = 3$, removable
4. $a = 3$
5. Yes, $\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^+} f(x) = 4$ and $f(2) \neq 4$, which makes this a removable discontinuity.
6. $a = -12$
7. $k = -3$
-  8. $b = 3.584$
9. D
10. C

AP Calc Lesson 1.9 Homework

Answer Key

1. Yes, since f is a continuous function, the Intermediate Value Theorem guarantees that $f(x)$ attains all values between $f(-2) = 8$ and $f(5) = 1$ on the interval $[-2, 5]$. Since $1 < 4 < 8$, there must be an x -value, $2 < x < 5$, such that $f(x) = 4$.
2. No, the number of annual visitors to Glacier National Park does not represent a continuous function. Values are discrete and “jump around”.
3.
 - a. Yes, temperature is a continuous function. It can't jump suddenly.
 - b. Since we can assume G to be continuous, the Intermediate Value Theorem guarantees that the temperature inside the greenhouse must attain all values between 18°C and 30°C between 8:00 AM and 11:00 AM and 25°C is between 18°C and 30°C .
4. Since f is a polynomial, f is a continuous function. The IVT guarantees that $f(x)$ attains all values between $f(0) = 2$ and $f(1) = -2$, which includes the value 0. Any x -value where $f(x) = 0$ represents an x -intercept on the graph of f .
5. 
 - a. $C = 35 + 0.12d(g)$
 - b. Yes, since d is a continuous function on the interval $[2, 5]$, the IVT guarantees that all distances between $d(2) = 48$ miles and $d(5) = 92$ miles are attained. Since the linear cost function is also continuous, the IVT also guarantees that all costs between $C(d(2)) = C(48) = \$40.76$ and $C(d(5)) = C(92) = \$46.04$ are attained on the interval $2 \leq g \leq 5$. Since $\$40.76 < \$42 < \$46.04$, there must exist a value of g in the interval $2 \leq g \leq 5$, for which the total cost of the rental is exactly $\$42$.

6.



7. a. No. At $x = 4$, $\lim_{x \rightarrow 4^-} g(x) = 4^2 - 10 = 6$ and $\lim_{x \rightarrow 4^+} g(x) = \sqrt{4+5} = 3$. Since $\lim_{x \rightarrow 4^-} g(x) \neq \lim_{x \rightarrow 4^+} g(x)$, g is not continuous at $x = 4$ due to a jump discontinuity and thus not continuous on $[0, 8]$.
- b. From part (a), we know g is not continuous on $[0, 8]$ due to a jump discontinuity at $x = 4$, so the IVT cannot be applied over the entire interval $[0, 8]$. However, we can still check whether $g(x) = 0$ has a solution in a subinterval where g is continuous. The function g is continuous on $[0, 4]$ since g is a quadratic polynomial. Since $g(0) = -10$ and $g(4) = 6$, the IVT guarantees there must be a value x for $0 \leq x \leq 4$ such that $g(x) = 0$. Since we are guaranteed such a value on $[0, 4]$, we are also guaranteed that value on $[0, 8]$.
-  8. a. Yes. Let $F(t)$ represent the overall rate of change of water in the tank, where $F(t) = R(t) - W(t)$. Since R and W are both continuous functions, F is also continuous. At time $t = 0$, $F(0) = R(0) - W(0) = 12$ liters per minute and at time $t = 10$, $F(10) = R(10) - W(10) = -1.612$ liters per minute. According to the IVT, there must be a time t on $[0, 10]$ at which $F(t) = 0$ since $-1.612 < 0 < 12$. When $F(t) = 0$, $R(t) = W(t)$ which means that the rate at which water is flowing into the tank is exactly the same as the rate at which water flows out of the tank.
- b. $t = 7.859$ minutes
9. B
10. B